

B.5. Qualifiers

Qualifiers are annotations that provide additional information regarding the use or interpretation of a schema construct. Often qualifiers are used to place restrictions on the form or range of values that the construct can assume.

B.5.1. any-order / schema-order / tag-order

Order Qualifiers Syntax

```
STRUCTURE [ any-order ]  
STRUCTURE [ schema-order ]  
STRUCTURE [ tag-order ]
```

The **any-order**, **schema-order** and **tag-order** qualifiers MAY be used to specify a particular order for the encoding of fields within a **STRUCTURE**.

The **any-order** qualifier specifies that the encoder of a TLV structure is free to encode the fields of the structure in any desired order.

The **schema-order** qualifier specifies that the fields of a structure SHALL be encoded in the order given within the associated **STRUCTURE** definition. If the **STRUCTURE** definition contains one or more **includes** statements, the fields of the referenced **FIELD GROUPs** SHALL be encoded in the order given in the respective **FIELD GROUP** definition, and at the position of the **includes** statement relative to other fields within the **STRUCTURE**.

The **tag-order** qualifier specifies that the fields of a structure SHALL be encoded in the order specified by their tags, as defined in [Section A.2.4, “Canonical Ordering of Tags”](#).

Only a single ordering qualifier MAY be applied to a given **STRUCTURE** type.

In the absence of an order qualifier, fields within TLV structure MAY generally be encoded in any order. However, the author of a **STRUCTURE** definition MAY choose to impose custom ordering constraints on some or all of the fields if so desired. Such constraints SHALL be clearly described in the prose documentation for the schema.

B.5.2. extensible

extensible Qualifier Syntax

```
STRUCTURE [ extensible ]
```

The **extensible** qualifier is only allowed on **STRUCTURE** types, and declares that the structure MAY be extended by the inclusion of fields not listed in its definition. When a structure is extended in this way, any new fields that are included SHALL use tags that are distinct from any of those associated with defined or included fields.

Absent the **extensible** qualifier, a structure encoding SHALL NOT include fields beyond those given